MATH 201 Your name: _____

Learning targets 1.3 and 1.5, version 1

Explain the meaning of the derivative in context. Assign the correct units to the value of the derivative.

Arapaho Glacier is a mountain glacier in Roosevelt National Forest, west of Boulder, CO. The following table[1] gives the surface area, $A(t)$, in square meters, of Arapaho Glacier in the year t.

t	1900	1960	1973	1999
$A(t)$	338,282	250,764	225,000	162,027

1. Compute an approximation for $A'(1960)$, and **include units** for this number.

2. Write a sentence explaining what the number means about how the area of the glacier is changing. **Don't say "per," and don't say "rate."**

3. Compute an approximation for $A'(1999)$, and **include units** for this number.
   Do you think your approximation is too high or too low? Why?

4. How does $A'(1960)$ compare to $A'(1999)$? Is that good or bad?

[1] Haugen, B., Scambos, T., Pfeffer, T., & Anderson, R. (2010). Twentieth-century changes in the thickness and extent of Arapaho Glacier, Front Range, Colorado. *Arctic, Antarctic, and Alpine Research, 42*(2), 198-209.

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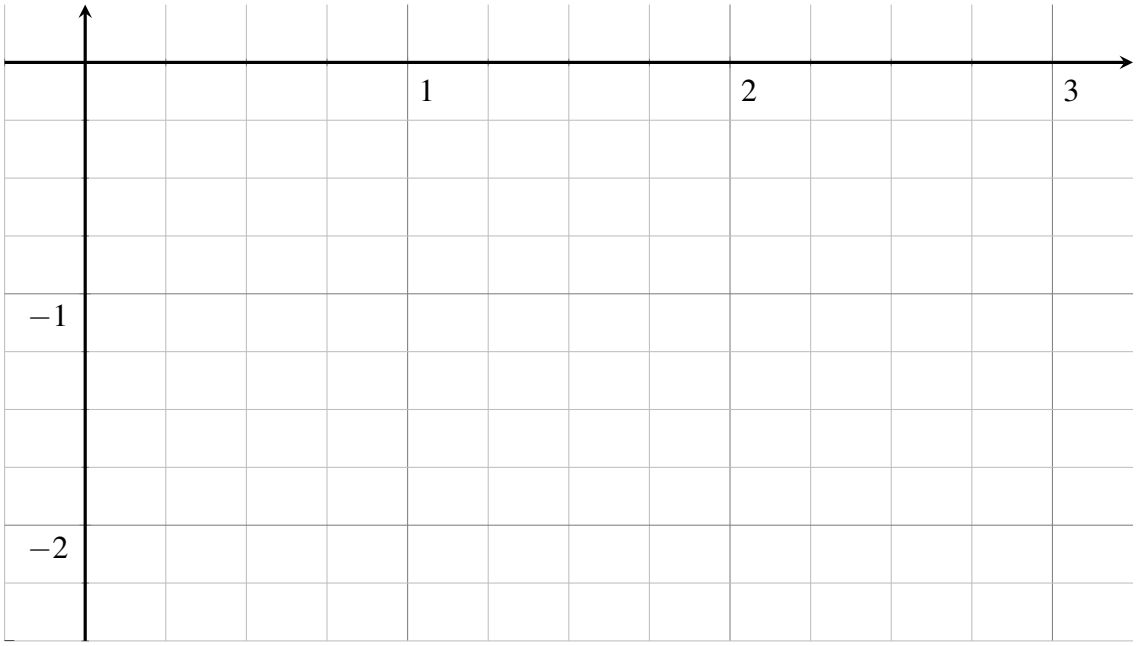
Learning target 1.3, version 2

Estimate the derivative of a function at a given value, using numerical or graphical data.

Suppose that you know the following values of some function $g(x)$:

x	1.7	2	2.3
$g(x)$	-2.2	-2	-1.5

1. Plot these points and sketch a plausible graph of $g(x)$.



2. On your graph above, draw a plausible tangent line to the graph of $g(x)$ at $x = 2$.

3. Use a central difference to estimate $g'(2)$. Draw the corresponding secant line on your graph above.

4. Compute another, different estimate of $g'(2)$. Draw the corresponding secant line on your graph.

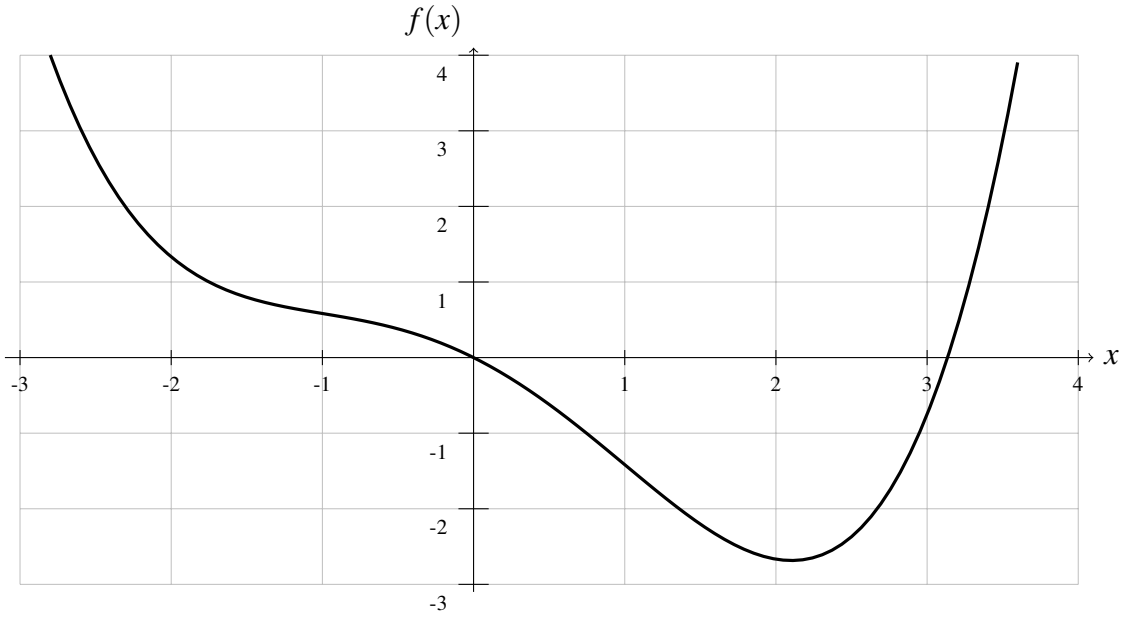
5. Which one of your approximations is the best? How do you know?

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Learning target 1.3, version 3

Estimate the derivative of a function at a given value, using numerical or graphical data.

Below is a graph of the function $f(x) = \dfrac{x^4}{12} - \dfrac{x^2}{2} - x$.



(a) On the graph above, draw the tangent line to the curve at $x = -1$.

(b) On the graph above, draw a secant line that goes through the curve at $x = -1$ and at some other nearby x-value of your choice.

(c) Calculate the slope of that secant line.

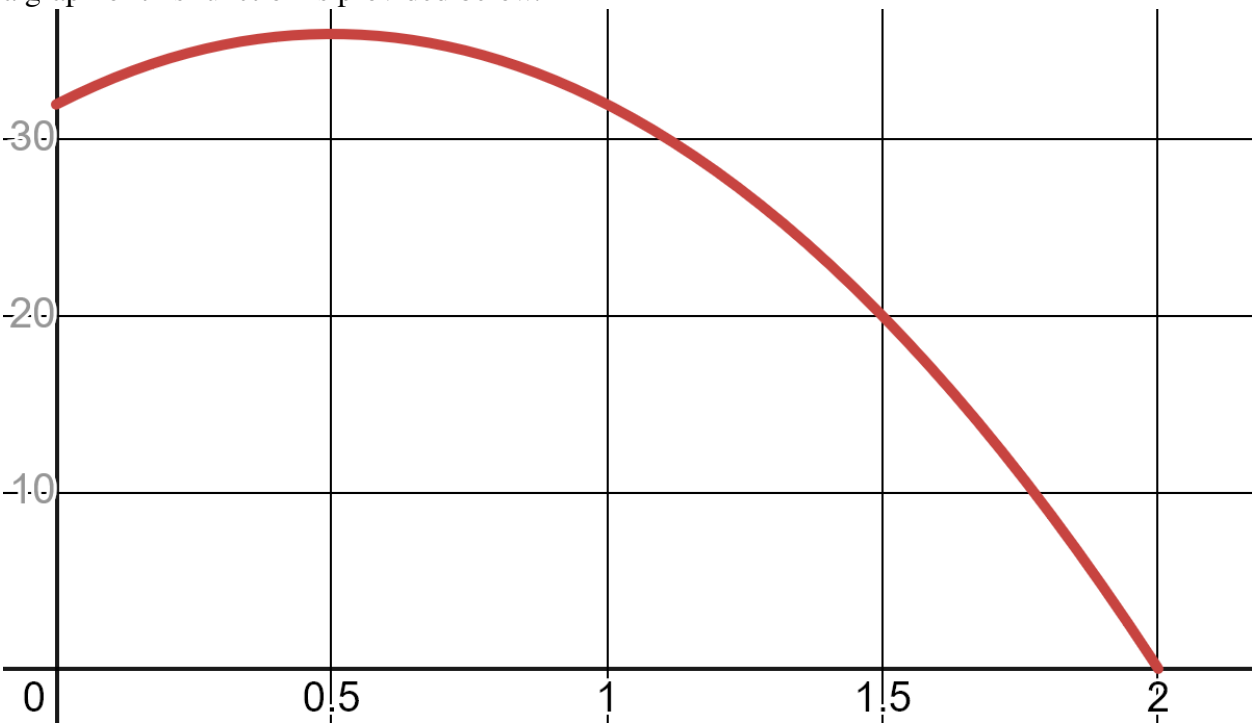
(d) Use $f'(x)$ to calculate the slope of the tangent line. Compare the value you get here to the value you got in part (c). Does that make sense?

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Learning target 1.3, version 4

Estimate the derivative of a function at a given value, using numerical or graphical data.

A water balloon is tossed vertically in the air from a window. The balloon's height (measured in feet) at time t (measured in seconds after being launched) is given by $s(t) = -16t^2 + 16t + 32$; a graph of this function is provided below.



(a) Compute $s(1)$. Show your work.

(b) On the graph above, carefully sketch the *tangent* line to $s(t)$ at $t = 1$.

(c) On the graph above, carefully sketch a *secant* line through the point on the graph at $t = 1$ and a second nearby point.

(d) Compute the slope of your secant line. Show your work and **include units**.

Hints: rise over run; one of your two y-values is $s(1)$, which you computed in part (a).

(e) Use shortcut rules to find $s'(t)$, and compute $s'(1)$; show your work. Compare your result here to your result in part (d). Do these two numbers make sense together? Why?

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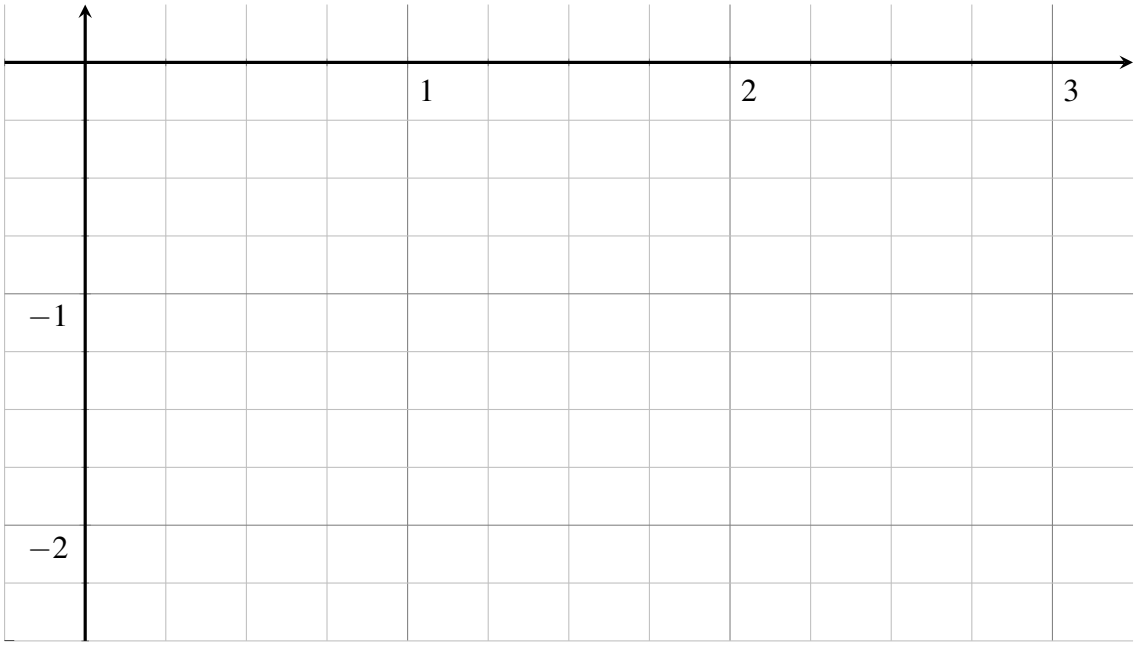
Learning target 1.3, version 5

Estimate the derivative of a function at a given value, using numerical or graphical data.

Suppose that you know the following values of some function $f(x)$:

x	1	1.5	2
$f(x)$	-0.8	-1	-1.6

1. Plot these points and sketch a plausible graph of $f(x)$.



2. On your graph above, draw a plausible tangent line to the graph of $f(x)$ at $x = 1.5$.

3. Use a central difference to estimate $f'(1.5)$. Draw the corresponding secant line on your graph.

4. Compute another, different estimate of $f'(1.5)$. Draw the corresponding secant line on your graph.

5. Which one of your approximations is the best? How do you know?